

UTS Cloud Computing Class - Lab Guide

Week 3 Lab: iSCSI (VMFS) on ESXi

If you have not finished the tasks of Week 2 Lab, please continue.

In Week 3, the lab tasks will connect your ESXi hosts to existing iSCSI. The storage appliance is already set up, so the configuration requirements are host related only. ***Note:** Not all virtual machines need to be accessed to complete this lab.

Task 1. Prepare for the lab.

1. Open VMware Workstation, and **power on** the following three VMs: 'ESXi01', 'ESXi02', and 'iSCSI'.
2. From the Mozilla Firefox browser, connect to the servers ESXi01 and ESXi02 using the URL. If not working, use the same method as last week to fix the network connections.
** Hint: Steps under 'To configure IP address', Week 2*
3. Verify that there is already a vm on each host from last week.
4. Change the IP address of iSCSI to 172.20.20.50, add mask 255.255.255.0, gateway, DNS as last week.

*** STOP *** If you are missing a vm, please go back to Task 5 from last week and follow the instructions to create the vms on each host.

Task 2. Record current datastores for the vms and the hosts.

1. Navigate to the **summary** page of a vm that you created last week. Record what datastore the vm is currently located on here _____.
**Hint: Click on 'vm1' on ESXi01 in the left panel -> Summary Page, Hardware Configuration -> Hard Disk 1 -> Backing [Datastore1]*
2. Repeat step 1. for your other vm on ESXi02. Record the datastore here _____.
3. For ESXi01, click 'Host' on left panel to navigate to the Summary page. Navigate to the **Hardware section**, then select the **Storage** option. How many datastores are listed?

4. Repeat step 3. for ESXi02 and record how many datastores are listed?

Task 3. Set up iSCSI connectivity for both hosts.

Open iSCSI in VMware Workstation from your student ID folder. Your iSCSI should be able to work directly.

1. From browser, ESXi01, select the **Storage** link on the left side panel.

2. Navigate to **Adapters** tab and select 'Software iSCSI'.
3. Scroll down to locate Dynamic targets section and click Add dynamic target. Click to add IP address as '**172.20.20.50**'.
4. Click 'Save Configurations'. Click **Rescan**.
5. Navigate to **Devices** tab and 'FreeNAS iSCSI Disk' should now appear in the list as well as should appear on the left panel under 'Storage'. Click 'Refresh' if it does not appear.
6. Repeat steps 1-5 for ESXi02.

Task 4. Verify iSCSI and create a datastore

1. From the left panel of ESXi01, click on the **Storage** link.
2. Is there another datastore available now that was not visible earlier? _____
3. Click **New Datastore** and select 'Create new VMFS datastore'. If there is a new Datastore, then delete the current datastore (you may need to delete/unregister all virtual machines that are attached to the current datastore).
4. Name the new datastore *YourName-vmfs1* and select the available device FreeNAS iSCSI Disk.
5. Check that the datastore you just created appears under the **Storage, Datastores** tab on ESXi02. ** If it does not appear, Refresh the browser. It take a number of minutes.*

Task 5. Prepare for next Lab

1. Right click each host, and select option **Shutdown**
2. Verify no virtual machines are running.
3. Shutdown the hosts ESXi01, ESXi02, and iSCSI VMs.

Note:

Make sure you take the backup of the VMs in your portable disk and delete the files from the folder on the lab computer.